

A Flux-Driven Compact-to-Continuous Transition for a Magnetic Grushin Cylinder

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Abstract

We give a complete channel-level spectral atlas for the magnetic Baouendi–Grushin form on $\mathbb{R} \times S^1$. The closed nonnegative form defines its Friedrichs realization, and Fourier–Hermite reduction gives every nonresonant level $(2n + 1)|k + \alpha|$.

1 Friedrichs realization and conventions

Use $L^2(\mathbb{R} \times S^1, dx d\theta / (2\pi))$ and write $D_\theta = -i\partial_\theta$. For $\alpha \in \mathbb{R}$, begin with

$$q_\alpha[u] = \int_{\mathbb{R} \times S^1} (|\partial_x u|^2 + x^2 |(D_\theta + \alpha)u|^2) \frac{dx d\theta}{2\pi}. \quad (1)$$

The form domain is the completion of compactly supported smooth functions in $\|u\|_2^2 + q_\alpha[u]$. Since (1) is densely defined, nonnegative, and closable, its closure determines a unique nonnegative self-adjoint operator by the representation theorem. We call it G_α . Formally,

$$G_\alpha = -\partial_x^2 + x^2(D_\theta + \alpha)^2.$$

This is a Friedrichs-form statement. We do not assert essential self-adjointness of a smaller differential core.

Boscain, Prandi, and Seri study spectral and magnetic effects on related almost-Riemannian Grushin manifolds [1]; their geometric Laplace–Beltrami realization is not silently identified with the present Lebesgue-space form. Harakeh and Hillairet analyze Fourier-separated Baouendi–Grushin cylinder sectors and compact-resolvent mechanisms [2]. Our theorem is a self-contained source-model closure, not a claim that these classical mechanisms are newly discovered.

2 Fourier–Hermite reduction

The unitary angular Fourier expansion $u(x, \theta) = \sum_{k \in \mathbb{Z}} u_k(x) e^{ik\theta}$ gives

$$G_\alpha \cong \bigoplus_{k \in \mathbb{Z}} h_{k+\alpha}, \quad h_\omega = -\frac{d^2}{dx^2} + \omega^2 x^2. \quad (2)$$

For $\omega \neq 0$, scaling the Hermite basis yields

$$\lambda_{k,n}(\alpha) = (2n + 1)|k + \alpha|, \quad k \in \mathbb{Z}, \quad n \in \mathbb{N}_0. \quad (3)$$

For $\omega = 0$, $h_0 = -d^2/dx^2$ is the free line Laplacian.

Theorem 1 (Complete flux dichotomy). *If $\alpha \notin \mathbb{Z}$, then G_α has compact resolvent and pure point spectrum given by (3), with multiplicities obtained by counting equal index pairs.*

If $\alpha \in \mathbb{Z}$, exactly one resonant angular Fourier channel is unitarily equivalent to the free line Laplacian. It contributes absolutely continuous spectrum $[0, \infty)$ of almost-everywhere multiplicity two. The orthogonal nonresonant sector has compact resolvent and point spectrum equal to every positive integer. Those eigenvalues are embedded in the full continuum. The singular-continuous spectrum is empty.

Proof. If $\delta = \text{dist}(\alpha, \mathbb{Z}) > 0$, every channel is an oscillator and (3) is complete. For fixed L , the inequality $(2n+1)|k+\alpha| \leq L$ permits only finitely many k and, for each k , only finitely many n . Thus the eigenvalue list has finite multiplicities and tends to infinity, which is equivalent to compact resolvent.

At integer flux, a gauge shift reduces to $\alpha = 0$. The $k = 0$ summand in (2) is the free line Laplacian. Fourier transformation in x shows that its spectrum is purely absolutely continuous on $[0, \infty)$; the two momentum branches $\xi = \pm\sqrt{E}$ give multiplicity two for almost every $E > 0$. Every $k \neq 0$ summand is an oscillator, and the same finite-list criterion makes their direct sum compact-resolvent. A direct sum of this pure-point part and the free absolutely continuous part has no singular-continuous component. Formula (3) becomes $(2n+1)|k|$, so its values are positive integers and lie inside the free continuum. This proves coexistence, rather than incorrectly replacing the full integer-flux spectrum by either component alone. $\square$

Flux shifts $\alpha \mapsto \alpha + j$, $j \in \mathbb{Z}$, merely relabel k . Complex conjugation and $k \mapsto -k$ give reflection $\alpha \mapsto -\alpha$. Hence the spectral fundamental interval is $0 \leq \delta \leq 1/2$.

References

- [1] U. Boscain, D. Prandi, and M. Seri, “Spectral analysis and the Aharonov–Bohm effect on certain almost-Riemannian manifolds,” *Commun. Partial Differential Equations* 41 (2016), 32–50, [arXiv:1406.6578](#).
- [2] M. Harakeh and L. Hillairet, “A spectral condition for the control of eigenfunctions of Baouendi–Grushin type operators,” [arXiv:2312.04359](#) (2023).